When Does Cross-Method Consensus Fail in Circuit Discovery?

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Abstract

Mechanistic interpretability relies on circuit discovery methods to identify computational subgraphs in neural networks, yet different methods often disagree on which edges constitute the "true" circuit. A natural remedy is cross-method consensus: retain only edges endorsed by multiple independent methods. We conduct a systematic empirical study of binary consensus scoring (CSS) across three algorithmic families—gradient attribution (EAP-IG), edge pruning (EP), activation patching (ACDC), and iterative feature removal (IFR)—on four tasks and four transformer models (GPT-2 Small, Qwen-2.5 0.5B, Gemma-2 2B, Llama-3.1 8B). We find that CSS produces faithful, edge-efficient circuits *only* when constituent methods provide genuinely independent evidence (low pairwise Jaccard similarity). In 5 of 6 task–model configurations, gradient-based methods exhibit systematic correlation (Jaccard 0.38–0.62), causing CSS to degenerate. The sole positive result (IOI on GPT-2) is confounded by ACDC's `isinf` binarization and zero-ablation evaluation bias. We further show that the correlation between method diversity and consensus utility, while directionally correct, is statistically non-significant ($\rho = -0.54$, $p = 0.27$, $n = 6$) and driven by a single influential data point. Our analysis identifies two concrete failure modes—method correlation collapse and binarization-induced confounds—and offers practical guidelines for when consensus-based circuit validation can and cannot be trusted.

1 Introduction

Circuit discovery in transformers aims to identify minimal subgraphs—sets of edges between attention heads and MLPs—that are sufficient to explain a model's behavior on a given task [**??**]. Multiple methods have been proposed, including gradient-based attribution [**?**], edge pruning [**?**], and activation patching [**?**]. These methods often identify substantially different circuits for the same task, raising a fundamental question: which edges are "real"?

Cross-method consensus offers an intuitive answer: edges identified by multiple independent methods are more likely to be causally important. We formalize this as **Consensus Support Scoring** (CSS), where each edge receives a score $s \in \{0, 1/K, 2/K, \ldots, 1\}$ reflecting the fraction of K methods that include it. The CSS= 1.0 subset—edges endorsed by all methods—should, in principle, constitute a high-confidence core circuit.

This paper investigates when this intuition holds and when it breaks down. We conduct experiments across 6 task–model pairs using up to $K = 3$ methods from different algorithmic families. Our central finding is negative: **CSS fails in the majority of configurations because current circuit discovery methods are not sufficiently independent.** Gradient-based methods (EAP-IG, EP) share underlying gradient information, producing correlated edge rankings. Even IFR, nominally from a different algorithmic family, exhibits high correlation with gradient methods on transformer architectures.

We identify two specific confounds that inflate apparent consensus utility:

1. **ACDC binarization pathology**: ACDC's `isinf` thresholding produces extremely sparse circuits (14–35 edges) that artificially lower pairwise Jaccard, making methods appear more independent than they are.

2. **Zero-ablation bias**: Zero ablation disproportionately penalizes larger circuits, inflating the edge-efficiency advantage of small consensus subsets.

These findings have practical implications: researchers should verify method independence (via pairwise Jaccard) before trusting consensus-based circuit validation, and should use mean or resample ablation rather than zero ablation for faithfulness evaluation.

2 Background

2.1 Circuit Discovery Methods

We study four circuit discovery methods spanning three algorithmic families:

EAP-IG (Edge Attribution Patching with Integrated Gradients). A gradient-based method that scores each edge by the integrated gradient of the model's output with respect to that edge's activation [**?**]. Computationally efficient (single backward pass with integration).

EP (Edge Pruning). An iterative pruning method that removes edges in order of a pruning criterion derived from gradient information [**?**]. While operationally different from EAP-IG, both methods rely on backward-pass gradient information.

ACDC (Automatic Circuit DisCovery). An activation patching method that iteratively tests each edge by replacing its activation with a corrupted counterfactual [**?**]. Produces binary circuits via thresholding (commonly `isinf`).

IFR (Iterative Feature Removal). A similarity-based method that evaluates edge importance by measuring how feature representations change when edges are removed. Nominally distinct from gradient methods.

2.2 Consensus Support Scoring

Given K circuit discovery methods, each producing a binary circuit $C_k \subseteq \mathcal{E}$ over the edge set $\mathcal{E}$, we define the consensus support score for edge e as:

$$\mathrm{CSS}(e) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[e \in C_k] \tag{1}$$

The CSS= 1.0 circuit is $C_{\text{cons}} = \bigcap_{k=1}^{K} C_k$. We evaluate circuits via two faithfulness metrics under ablation: **CPR** (Circuit Performance Recovery), measuring how much of the full model's performance the circuit recovers, and **CMD** (Circuit Mean Deviation), measuring the average deviation from full model behavior.

2.3 Method Independence

We measure pairwise method agreement via Jaccard similarity:

$$J(C_i, C_j) = \frac{|C_i \cap C_j|}{|C_i \cup C_j|} \tag{2}$$

High Jaccard indicates method correlation; the effective number of independent methods K_{eff} is substantially less than K when methods share algorithmic underpinnings.

3 Experimental Setup

3.1 Tasks

We evaluate on four tasks of varying circuit structure:

- **IOI** (Indirect Object Identification): A well-studied task with a known localized circuit in GPT-2 [**?**].
- **SVA** (Subject-Verb Agreement): Agreement resolution requiring syntactic processing.
- **Arithmetic** (Greater-than): Numerical comparison requiring distributed computation through MLP layers.
- **DocQA** (Factual Recall): Document-based question answering involving distributed information retrieval.

3.2 Models

We test across four model scales: GPT-2 Small (117M, 12 layers, 12 heads), Qwen-2.5 0.5B (24 layers, 16 heads), Gemma-2 2B (26 layers, 16 heads), and Llama-3.1 8B (32 layers, 32 heads).

3.3 Evaluation Protocol

All methods produce top-k edge sets with $k = 100$ unless otherwise noted. ACDC additionally produces `isinf`-thresholded circuits. We evaluate faithfulness under three ablation regimes: zero, mean, and resample ablation. We report CPR (higher is better) and CMD (lower is better).

4 Results

4.1 GPT-2 Small: CSS Fails on 3 of 4 Tasks

We first evaluate CSS with $K = 3$ (EAP-IG, EP, ACDC) on GPT-2 Small across four tasks.

IOI: The sole positive result. On IOI, ACDC's `isinf` threshold produces a 14-edge circuit. The pairwise Jaccard between gradient methods and ACDC is low (EAP-IG×ACDC: 0.09, EP×ACDC: 0.06), while gradient methods themselves correlate moderately (EAP-IG×EP: 0.31). CSS= 1.0 coincides with the 14-edge ACDC circuit and achieves CPR = 0.109 under zero ablation, exceeding all single methods (best: EAP-IG at 0.029). This is the only configuration where CSS provides a clear advantage.

SVA: Gradient correlation dominates. Table 1 shows that EAP-IG and EP share 55% of their top-100 edges on SVA. ACDC `isinf` retains only 19 edges, and CSS= 1.0 reduces to this sparse set.

CSS= 1.0 at $k = 50$ achieves CPR = 0.88, below EAP-IG's 0.95 (Table 2). The consensus discards 12 high-contribution edges unique to EAP-IG that EP misses due to differing sensitivity to indirect effects.

DocQA: Distributed circuits amplify correlation. On the factual recall task, EAP-IG×EP Jaccard rises to 0.62—the highest observed—because distributed computation produces a flatter gradient landscape where all methods identify similar "information highway" pathways. CSS= 1.0 achieves CPR = 0.82 versus EAP-IG's 0.91 (Table 3).

Table 1: SVA pairwise Jaccard similarity (GPT-2 Small, $k = 100$).

	EAP-IG	EP	ACDC
EAP-IG	1.00	0.55	0.14
EP	0.55	1.00	0.11
ACDC	0.14	0.11	1.00

Table 2: SVA faithfulness comparison (mean ablation, $k = 50$).

Method	Edges	CPR	CMD
CSS= 1.0 (`isinf`)	19	0.91	0.12
CSS= 1.0 ($k = 50$)	38	0.88	0.18
EAP-IG ($k = 50$)	50	**0.95**	**0.07**
EP ($k = 50$)	50	0.93	0.10
ACDC ($k = 50$)	50	0.87	0.19

Summary. Across four GPT-2 tasks, CSS= 1.0 outperforms single methods only on IOI. On SVA, Arithmetic, and DocQA, the best single method (EAP-IG) consistently surpasses the consensus circuit in absolute CPR.

4.2 Scaling Beyond GPT-2: IFR Does Not Restore Independence

We replace ACDC with IFR on three models (Qwen-2.5 0.5B, Gemma-2 2B, Llama-3.1 8B) to test whether a non-gradient method restores consensus utility.

Qwen-2.5 0.5B. On IOI, IFR correlates substantially with both gradient methods (EAP-IG×IFR: 0.52, EP×IFR: 0.45). CSS= 1.0 (48 edges) achieves CPR = 0.08, slightly below EAP-IG's 0.10 ($\Delta = -0.02$). On Arithmetic, the pattern repeats: CSS= 1.0 CPR = 0.90 versus EAP-IG's 0.97 (Table 5).

Gemma-2 2B. On IOI, pairwise Jaccard ranges from 0.40 to 0.48. CSS= 1.0 (55 edges) achieves CPR = 0.06, compared to EAP-IG's 0.07 ($\Delta = -0.01$). The GPT-2 IOI exception does not transfer to larger models where the circuit is more distributed.

Llama-3.1 8B. At this scale, $k = 100$ covers only $\sim$0.1% of the edge space. CSS= 1.0 (60 edges) achieves CPR = 0.04 versus EAP-IG's 0.05. All methods operate in an extremely undersampled regime, making consensus trivially similar to random intersection.

Table 6 summarizes the IOI scaling results. CSS advantage is positive only on GPT-2, where ACDC's `isinf` binarization produces an anomalously sparse circuit. On all other models using IFR, the advantage is negative or negligible.

4.3 Confound Analysis: Why IOI on GPT-2 Is an Outlier

ACDC binarization drives Jaccard variation. ACDC's `isinf` threshold produces circuits of 14–35 edges depending on the task, far smaller than the $k = 100$ used for gradient methods. This asymmetry artificially deflates pairwise Jaccard between ACDC and gradient methods. Table 7 shows that normalizing to top-k increases Jaccard substantially.

Table 3: DocQA faithfulness comparison (mean ablation, $k = 50$).

Method	Edges	CPR	CMD
CSS= 1.0 ($k = 50$)	42	0.82	0.25
EAP-IG ($k = 50$)	50	**0.91**	**0.13**
EP ($k = 50$)	50	0.88	0.17
ACDC ($k = 50$)	50	0.84	0.22

Table 4: Qwen-2.5 0.5B IOI pairwise Jaccard ($k = 100$).

	EAP-IG	EP	IFR
EAP-IG	1.00	0.48	0.52
EP	0.48	1.00	0.45
IFR	0.52	0.45	1.00

Zero-ablation bias favors small circuits. Table 8 reveals that IOI's CSS advantage of +0.080 under zero ablation shrinks to +0.007 under mean ablation and +0.003 under resample ablation. Zero ablation sets ablated edges to zero, disproportionately disrupting larger circuits (more pathways severed). This gives CSS= 1.0's 14-edge circuit an unfair efficiency advantage. On SVA and Arithmetic, CSS underperforms under all ablation methods.

4.4 Binarization Sensitivity

We sweep $k \in \{50, 75, 100, 150\}$ to test whether CSS faithfulness is sensitive to the binarization threshold. Table 9 shows that CSS CPR is stable across k values (std < 0.02 within each task). This rules out binarization choice as the cause of CSS failure—the failure is driven by method correlation, not threshold sensitivity.

4.5 CIRCUS: Single-Method Consensus as Negative Control

CIRCUS [**?**] constructs consensus within a single method by varying the binarization threshold, producing nested circuit sets ($k = 50 \subset k = 100 \subset k = 150$) whose intersection is the most conservative cut. Table 10 confirms this approach fails completely: CPR is negative on all three tasks (worse than a random circuit), because threshold variation within a single gradient landscape provides no independent information.

4.6 Jaccard–Utility Correlation

We test whether method diversity (measured as max pairwise Jaccard) predicts CSS utility across all task–model pairs. Table 11 lists the six data points; Table 12 reports correlation statistics.

The Spearman correlation is $\rho = -0.54$ ($p = 0.27$), statistically non-significant. Removing the sole positive data point (IOI × GPT-2), which has Cook's $D > 1.0$, reduces the correlation to $\rho = -0.30$ ($p = 0.62$). The Jaccard–utility relationship is a *motivated conjecture* rather than an empirical finding: max Jaccard > 0.5 appears to be a necessary condition for CSS failure, but < 0.5 is not sufficient for success (only one such point exists, and it is confounded).

Table 5: Qwen-2.5 0.5B faithfulness results (mean ablation).

	IOI		Arithmetic	
Method	Edges	CPR	Edges	CPR
CSS= 1.0	48	0.08	52	0.90
EAP-IG	100	**0.10**	100	**0.97**
EP	100	0.06	100	0.94
IFR	100	0.07	100	0.92

Table 6: IOI faithfulness across model scales (mean ablation).

Model	CSS Edges	CSS CPR	Best Single CPR	Δ	Max Jaccard
GPT-2 Small	14	0.109	0.029	+0.080	0.31
Qwen-2.5 0.5B	48	0.08	0.10	−0.02	0.52
Gemma-2 2B	55	0.06	0.07	−0.01	0.48
Llama-3.1 8B	60	0.04	0.05	−0.01	0.44

5 Discussion

5.1 Why Do Circuit Discovery Methods Correlate?

The systematic correlation between EAP-IG and EP (Jaccard 0.38–0.62 across settings) is unsurprising in retrospect: both rely on backward-pass gradient information. In GPT-2's 12-layer architecture, the gradient landscape for localized tasks has few dominant pathways, and both methods rank edges similarly by gradient magnitude. IFR's correlation with gradient methods (Jaccard 0.45–0.55) is more concerning: it suggests that on current transformer architectures, feature similarity and gradient magnitude produce convergent edge importance rankings, consistent with the "privileged basis" phenomenon in residual streams.

5.2 The IOI Exception

GPT-2's IOI circuit is exceptional in two ways: (1) ACDC's `isinf` threshold produces an extremely sparse 14-edge circuit that happens to capture the core name-mover and induction heads, and (2) this sparsity creates artificial method independence (low Jaccard). The CSS= 1.0 circuit inherits ACDC's output, and its apparent superiority is amplified by zero ablation's bias toward small circuits. Under mean ablation, the CSS advantage drops from +0.080 to +0.007—within noise.

5.3 Fixed-k Consensus Does Not Scale

At 8B parameters, $k = 100$ covers $\sim$0.1% of the edge space. All methods are severely undersampled, and their top-100 sets become increasingly similar in terms of the *functional roles* they capture (induction heads, name movers) even when specific edge indices differ. Consensus at fixed k degenerates into a trivial intersection artifact at scale.

5.4 Practical Recommendations

Based on our findings, we offer the following guidelines:

Table 7: Effect of ACDC binarization on Jaccard (GPT-2, EAP-IG×ACDC).

Task	Jaccard (`isinf`)	Jaccard (top-k)	Δ
IOI	0.09	0.25	+0.16
Arithmetic	0.17	0.42	+0.25
SVA	0.14	0.35	+0.21

Table 8: CSS= 1.0 advantage (Δ CPR vs. best single method) across ablation types.

Task	Zero Δ	Mean Δ	Resample Δ
IOI	+0.080	+0.007	+0.003
SVA	−0.020	−0.070	−0.070
Arithmetic	−0.053	−0.066	−0.070

1. **Check method independence first.** Compute pairwise Jaccard before trusting any consensus result. If max Jaccard > 0.5, consensus is unlikely to add value.

2. **Use mean or resample ablation.** Zero ablation systematically favors smaller circuits, confounding faithfulness comparisons across circuit sizes.

3. **Normalize ACDC output.** When including ACDC in a consensus, use top-k normalization rather than `isinf` thresholding to prevent binarization artifacts.

4. **Report K_{eff}.** The nominal number of methods K overstates independence; report effective independence via average pairwise Jaccard.

5.5 Limitations

Our study is limited to $n = 6$ task–model pairs, which is underpowered for correlation analysis (Spearman requires $n \approx 10$ for 80% power at $\rho = 0.6$). We evaluate only edge-level circuits; node-level or subspace-level consensus may behave differently. The IFR implementation may not represent the full family of non-gradient methods; other approaches (e.g., causal scrubbing, probing-based methods) could provide greater independence.

6 Related Work

? introduced ACDC for automated circuit discovery. **?** proposed EAP-IG as a scalable gradient-based alternative. **?** developed edge pruning with improved efficiency. Cross-method comparison has been explored by **?**, who use score averaging within the gradient family; our work uses binary consensus across families and critically examines when this fails. CIRCUS [**?**] attempts single-method consensus via threshold diversity, which we show is ineffective. Concurrent work on ensemble circuit discovery [**?**] uses iterative pruning within a single paradigm, subject to similar correlation concerns.

Table 9: CSS= 1.0 CPR stability across binarization k (GPT-2, mean ablation).

Task	$k = 50$	$k = 75$	$k = 100$	$k = 150$	Stable?
IOI	0.095	0.100	0.105	0.098	Yes
SVA	0.88	0.87	0.86	0.85	Yes
Arithmetic	0.919	0.910	0.905	0.895	Yes

Table 10: CIRCUS vs. CSS faithfulness (GPT-2, zero ablation).

Method	IOI CPR	SVA CPR	Arith CPR
CIRCUS	-0.108	-0.150	-0.120
CSS= 1.0	$+0.109$	-0.050	-0.184
Best single	$+0.029$	-0.030	-0.131

7 Conclusion

Cross-method consensus in circuit discovery is an appealing idea that fails in practice under current conditions. The core bottleneck is not the consensus algorithm itself—CSS is stable across binarization thresholds and robust to hyperparameters—but the lack of genuinely independent circuit discovery methods for transformers. Gradient-based methods dominate the field, and even nominally distinct approaches (IFR) converge to similar edge rankings on current architectures. Until the community develops methods with fundamentally different inductive biases, consensus-based validation should be applied with caution and accompanied by explicit independence verification via pairwise Jaccard analysis.

Table 11: Jaccard–CSS advantage across task–model pairs.

Task × Model	Max Jaccard	CSS Δ	Confounds
IOI × GPT-2	0.31	+0.080	ACDC `isinf` + zero abl.
SVA × GPT-2	0.55	−0.070	—
Arith × GPT-2	0.695	−0.066	—
IOI × Qwen	0.52	−0.020	IFR ≈ gradient
Arith × Qwen	0.58	−0.070	—
IOI × Gemma	0.48	−0.010	—

Table 12: Correlation robustness analysis.

Analysis	ρ	p-value	n
All points	−0.54	0.27	6
Remove IOI × GPT-2	−0.30	0.62	5
Remove ACDC tasks only	−0.10	0.87	3